Term Project: Deep Learning for Disaster Management

Overview

In this term project, you will apply deep learning to address a specific challenge in disaster management, a field critical for improving preparedness, response, and recovery from natural or human-induced disasters. You will select a sub-topic or task within this theme (e.g., predicting floods, detecting wildfire damage, or classifying disaster-related social media posts).

Specifically, you will develop a deep learning model using PyTorch or TensorFlow, work with a publicly available dataset, evaluate performance, write a 4-8 page paper, and present your findings/results. The project encourages creativity while ensuring feasibility, offering opportunities to explore neural networks such as CNNs, RNNs, and Transformers, and to address ethical considerations (e.g., model fairness in disaster scenarios). You will work in groups of up to 3 students to foster collaboration, with clear milestones to keep you on track.

Objectives

- **Apply deep learning techniques:** Design and implement a neural network tailored to your chosen disaster management task (e.g., classification, prediction, detection).
- **Work with real-world data:** Select and preprocess a public dataset, handling challenges like noise or imbalance.
- **Evaluate and optimize:** Use appropriate metrics (e.g., accuracy, F1-score, RMSE) and experiment with model variations.
- **Communicate findings:** Articulate your approach, results, and implications in a paper and presentation.

Timeline (6 Weeks)

- **Week 1 -- Project Proposal (Due by 11:59 pm, Oct 27)**
  - Each group needs to submit a 1-page proposal including:
    - **Sub-Topic and Title:** A specific, descriptive title (e.g., "Deep Learning for Flood Damage Detection in Aerial Imagery").
    - **Problem Statement:** Explain the disaster management challenge and its significance.
    - **Dataset:** Identify a publicly available dataset (include source/link; see examples below).
    - **Model Approach:** Outline your planned deep learning architecture (e.g., CNN, LSTM) and a baseline for comparison (e.g., logistic regression).
    - **Evaluation Metrics:** Specify metrics (e.g., precision, recall, MSE).
- **Week 2-3 Model Implementation & Initial Training**
  - **Intermediate Update 1 (by 11:59pm, Nov 10):** Submit a 1-2 page report and code checkpoint (via Github) showing:
    - Preprocessed dataset and exploratory analysis (e.g., visualizations of data distribution).
    - Initial model implementation and training results (e.g., loss curves, preliminary metrics).
    - Challenges faced and next steps.
- **Week 4-5 Experimentation, evaluation, and refinements**
  - **Intermediate Update 2 (by 11:59pm, Nov 24):** Submit a 1-2 page report and code checkpoint (via Github) showing:
    - Experimental results (e.g., hyperparameter tuning, architecture variations).
    - Evaluation metrics and comparisons with baseline.
    - Plans for final deliverables (paper, presentation).
- **Week 6 Paper Writing, Presentation Prep, Final Code Polish**

- **Project Showcase and Presentation (Dec 2)**
- **Project Code (Github)/Paper Due (Dec 5)**

Requirements

- **Dataset:** Must be publicly available, disaster-related, and suitable for deep learning. Avoid overly large or proprietary datasets. See suggested datasets below.
- **Model Development:** Implement a deep learning model in PyTorch or TensorFlow, starting with a simple architecture (e.g., ResNet-based transfer learning) and optionally extending to advanced methods (e.g., ViT).
- **Team Size:** Groups of up to 3 students are encouraged (individual projects allowed). This size balances collaboration and accountability, allowing the division of tasks such as data preprocessing, model coding, and evaluation while avoiding the coordination issues of larger groups.

Suggested Sub-Topics and Datasets

Choose a focused sub-topic within disaster management. Below are examples with tasks and public datasets to inspire your proposal. Ensure datasets are manageable.

- **Wildfire Detection and Spread Prediction:**
    - Task: Detect wildfires in satellite images or predict spread using environmental data (e.g., CNNs or RNNs).
    - Datasets:
        - Kaggle's Wildfire Detection Dataset (satellite images with fire hotspots).
        - UCI Forest Fires Dataset (tabular data for fire area regression).
- **Severe Weather Prediction:**
    - Task: Predict rainfall or storm intensity using time-series data (e.g., LSTMs).
    - Datasets:
        - UCI Air Quality Dataset (weather-related features for prediction).
        - Kaggle's Rainfall Prediction Dataset (historical weather data for regression).
- **Flood Damage Assessment:**
    - Task: Classify or segment aerial/satellite images for flood-affected areas (e.g., CNNs).
    - Datasets:
        - Kaggle's Flood Detection Dataset (satellite images labeled for flooding).
        - SpaceNet Flooding Challenge Dataset (aerial imagery for segmentation).
- **Earthquake Impact Analysis:**
    - Task: Classify building damage in images or predict aftershock magnitude (e.g., CNNs or MLPs).
    - Datasets:
        - xBD Dataset (satellite images for damage classification).
        - USGS Earthquake Dataset (time-series for magnitude prediction).
- **Disaster Response from Social Media:**
    - Task: Classify tweets for urgency or event type during disasters (e.g., Transformers).
    - Datasets:
        - CrisisNLP Dataset (labeled tweets for multi-class classification).
        - Kaggle's Natural Disaster Tweets Dataset (text for sentiment/event detection).
- **Hurricane/Typhoon Tracking:**
    - Task: Predict hurricane paths or intensity from meteorological data (e.g., LSTMs, CNNs).
    - Datasets:
        - NOAA HURDAT2 (time-series for path/intensity prediction).
        - Kaggle's Hurricane Damage Imagery (images for damage assessment).

Find datasets on Kaggle, UCI ML Repository, PapersWithCode, or Google Dataset Search. Verify dataset size and compatibility with your task in the proposal.

Deliverables and Grading (Total 300 Points)

- Project Proposal (50 Points)
  - Clarity and specificity of sub-topic/title (15 points).
  - Relevance to disaster management and motivation (15 points).
  - Feasibility of dataset and model plan (20 points).
- Intermediate Update 1 (50 Points)
  - Data preprocessing and exploration quality (20 points).
  - Initial model implementation and results (20 points).
  - Clarity of report and next steps (10 points).
- Intermediate Update 2 (50 Points)
  - Depth of experimentation (e.g., tuning, variations) (20 points).
  - Evaluation metrics and baseline comparison (20 points).
  - Report clarity and final deliverable plan (10 points).
- GitHub Repository (50 Points)
  - Clean, organized code for data, model, and evaluation (20 points).
  - Reproducibility (code runs without errors) (20 points).
  - README with clear instructions (10 points).
- Paper (50 Points):
  - Structure and clarity (introduction, methodology, etc.) (15 points).
  - Depth of analysis (results, discussion, ethics) (20 points).
  - Proper citations and formatting (15 points).
  - Format: 4-8 pages, conference-style (e.g., IEEE/ACM template).
  - Sections:
    - Introduction: Sub-topic, motivation, objectives.
    - Related Work: 3-5 cited papers on similar applications.
    - Methodology: Dataset, model architecture (with diagrams), training.
    - Experiments and Results: Metrics, visualizations, analysis.
    - Discussion: Limitations, ethical issues (e.g., bias in disaster response), future work.
    - References.
- Presentation (50 Points):
  - Clarity and engagement of slides (15 points).
  - Explanation of methodology and results (20 points).
  - Demo quality and Q&A handling (15 points).
  - Format: 8-10 minutes, slide deck (e.g., Google Slides) with demo (e.g., model predictions on sample inputs).

Tips for Success

- Choose a Manageable Scope: Text or tabular tasks (e.g., tweet classification) are lighter on computation than image tasks. Start simple and add complexity if time allows.
- Group Collaboration: In groups of 2-3, divide tasks (e.g., one for data, one for modeling, one for evaluation). Use GitHub for version control.
- Dataset Selection: Pick datasets with a reasonable size (e.g., <1GB). Check data quality (e.g., labels and sizes).

- Ethical Considerations: Discuss biases (e.g., models failing in underrepresented regions) to strengthen your paper.
- Checkpoints: Use intermediate updates to stay on track—aim for a working baseline.
- Resources: Leverage Colab for GPUs, Hugging Face for NLP models, and office hours for conceptual guidance (not code writing).

This project will build your skills in applying deep learning to impactful disaster management problems.

Notes on Group Size and Grading:

- **Group Size:** Groups of up to 3 are reasonable, as they allow task division (e.g., data preprocessing, model coding, evaluation) without excessive coordination overhead. Each member should contribute to all deliverables, with clear documentation in the GitHub README of individual contributions to ensure fair grading.
- **Grading Fairness:** The 300-point breakdown is balanced, with 100 points for intermediate progress (Updates 1 and 2), ensuring students stay on track. The proposal (50 points) encourages early planning, while the paper, presentation, and GitHub (50 points each) reward communication and technical quality.